Predicting Next Status: Model Experiments

Group 2 | Logistic Regression and Random Forest | Experiment report

Experimental design and evaluation

We compared logistic regression (LR) and random forest (RF) models for the three-class NEXT_STATUS target (0, 1, 2). The notebook imported at most 1,000,000 records, retained the supplied eligibility filters, and used a fixed 25%/75% random split with seed 12345. The resulting samples contained 233,185 training rows and 700,403 held-out rows. The 500,000-row fitting limit did not bind. All 12 models used the same data split and 14 supplied features, so differences reflect model settings rather than a different sample.

The pipeline assembles the original feature columns and standardizes them using training-fitted statistics. Although the notebook creates categorical encoders, its assembler prefers the original columns, so these experiments do not test one-hot encoding. We retained this behavior to focus on the requested classifier parameters. LR maxIter was fixed at 100; RF seed was fixed at 12345. Each experiment had a distinct saved name, and evaluation used reloaded models.

Our primary criterion was mean h_fg = mean[-ln(p_true)], the notebook's multiclass log loss; lower is better. It measures the probability assigned to the observed class and penalizes confident mistakes. The updated h_ug averages -ln(p_k) over positive class probabilities for each row, then averages across rows. Because h_ug does not use the actual label, we treated it as a probability-concentration diagnostic rather than a model-selection objective. All models had 700,403 non-null held-out h_fg values.

Parameter search and results

Experiment	LR reg.	LR L1 share	LR h_fg	RF trees	RF depth	RF h_fg
baseline	0.001	0.001	0.156826	12	6	0.159079
exp02	0.01	0.001	0.157432	24	6	0.158179
exp03	0.1	0.001	0.160680	48	6	0.158563
exp04	0.01	0.5	0.160172	24	4	0.162532
exp05	0.01	1.0	0.163487	24	8	0.155490
exp06	0.0001	0.001	0.156800	48	10	0.153104

Table 1. Held-out log loss on the common split. LR L1 share is elasticNetParam; LR reg. is regParam. Experiment names have the prefix group2_w2_.

For LR, we increased regularization to test whether shrinkage improved generalization, then increased the L1 share to test a sparser fit. Neither approach improved on baseline. At regParam = 0.01, moving the L1 share from 0.001 to 0.5 and 1.0 progressively worsened loss. We therefore tried weaker regularization in exp06, which gave the best LR score, but only a small improvement over baseline.

For RF, we tested additional trees at depth 6 to reduce ensemble variance, then varied depth with 24 trees to test capacity. Moving from 12 to 24 trees helped; 48 trees at depth 6 did not help further. Depth 4 underfit relative to depths 6 and 8. Exp06 combined 48 trees and depth 10 and achieved the lowest held-out loss. Since both parameters changed from exp05, the gain cannot be attributed to depth alone.

Feature-level diagnostics and distribution shift

We used the notebook's graphs to compare mean predicted probabilities with observed class frequencies across rounded feature-percentile groups. The plots describe associations within the held-out sample, not causal effects or calibration by predicted-probability bins. To summarize discrepancies, we calculated count-weighted absolute differences between each group's mean prediction and observed frequency, separately for classes 0 and 2.

Feature	exp05: class 0	exp06: class 0	exp05: class 2	exp06: class 2
AGE	0.005367	0.003976	0.001267	0.001003
CREDIT_SCORE	0.001771	0.001620	0.003527	0.002606
INCENTIVE	0.004855	0.003943	0.001255	0.001094
MTM_LTV	0.003338	0.002822	0.001321	0.001299

Table 2. Feature-group probability error on 700,403 held-out rows; lower is better. Each entry is sum(n_g |mean prediction_g - observed frequency_g|) / sum(n_g).

Exp06 had smaller group errors than exp05 for both inspected classes across AGE, CREDIT_SCORE, INCENTIVE, and MTM_LTV. This gives evidence beyond the aggregate loss that the improvement is spread across several feature views. However, a group average can hide individual errors and small subpopulations; these diagnostics do not establish robustness to a new distribution.

RF candidate	Training h_fg	Held-out h_fg	Held-out minus train
exp05: 24 trees, depth 8	0.152635	0.155490	0.002855
exp06: 48 trees, depth 10	0.143609	0.153104	0.009495

Table 3. The deeper candidate improves held-out loss but has a larger generalization gap.

The instructor's evaluation data come from a different distribution. A deeper forest can capture useful nonlinear relationships and interactions, but its larger gap raises concern about fitting patterns specific to the available training data. Exp05 is a credible lower-complexity alternative. A smaller gap alone does not prove that exp05 transfers better: its held-out loss and the inspected feature-group errors were worse. We chose to balance complexity against the consistent measured gains, while avoiding a claim that current-distribution validation predicts the final ranking.

Class imbalance also matters: the held-out frequencies were approximately 2.358% for class 0 and 0.915% for class 2, leaving 96.727% for class 1. Overall accuracy would obscure probability errors for rare outcomes, so we used log loss and inspected the two minority classes. Changes in class prevalence or feature relationships could alter model performance. Our random split is neither a temporal nor a distribution-shift test. Moreover, repeated tuning on this held-out split makes it a validation set, not an untouched final test; no multi-seed or uncertainty analysis was performed.

Selected submission and reproducibility

We submit **group2_w2_exp06_rf**, a random forest with **48 trees, maxDepth 10, and seed 12345**. Its held-out h_fg of **0.153104** is about 3.76% lower than baseline RF and 2.36% lower than the best LR. The stronger aggregate score and lower inspected feature-group errors support this selection, subject to the distribution-shift limitations above.

The submission includes the complete **group2_w2_exp06_rf_model** Spark PipelineModel directory and the exported Zeppelin notebook. The saved pipeline contains feature assembly, scaling, and the classifier. A reload check produced 1,000 valid three-class probability vectors. All notebook paths derive from the base folder interface field; group name **group2_w2_exp06** selects the candidate. Full workflow reproduction requires the original training data and a compatible Spark/Zeppelin environment; the saved model can be loaded independently for inference on compatible evaluation inputs.